Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 1 (Generated practice paper - Set 18)

Assessed: The physical environment, Energy resources, Pollution, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

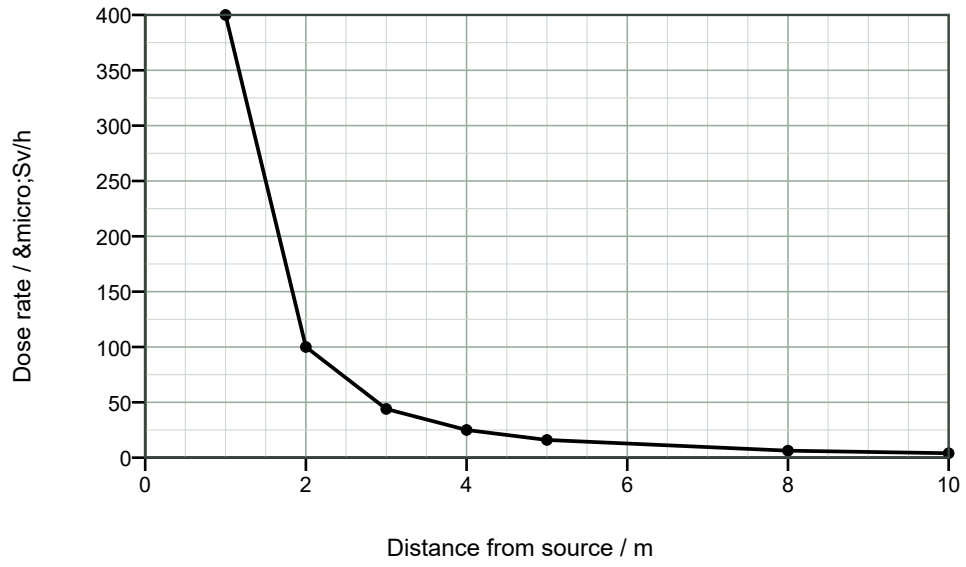
Question	Mark
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11	
TOTAL	

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Answer all questions in the spaces provided.

0 1 Figure 1 shows the radiation dose rate measured at different distances from a small gamma source.

Figure 1



0 1 . 1 Use Figure 1 to show that the dose rate follows the inverse square law.

[2 marks]

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0 1 . 2 Calculate the dose rate at 20 m from the source.

[1 mark]

Answer µSv/h

0 1 . 3 Explain how each of the following reduces the exposure of workers handling radioactive sources.

[4 marks]

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Closed (sealed) sources .....

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Absorbing materials .....

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Reduced exposure time .....

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Worker monitoring .....

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0 1 . 4 Explain why potassium iodide tablets are issued to people near a nuclear accident.

[2 marks]

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0 1 . 5 State the meaning of ALARA.

[1 mark]

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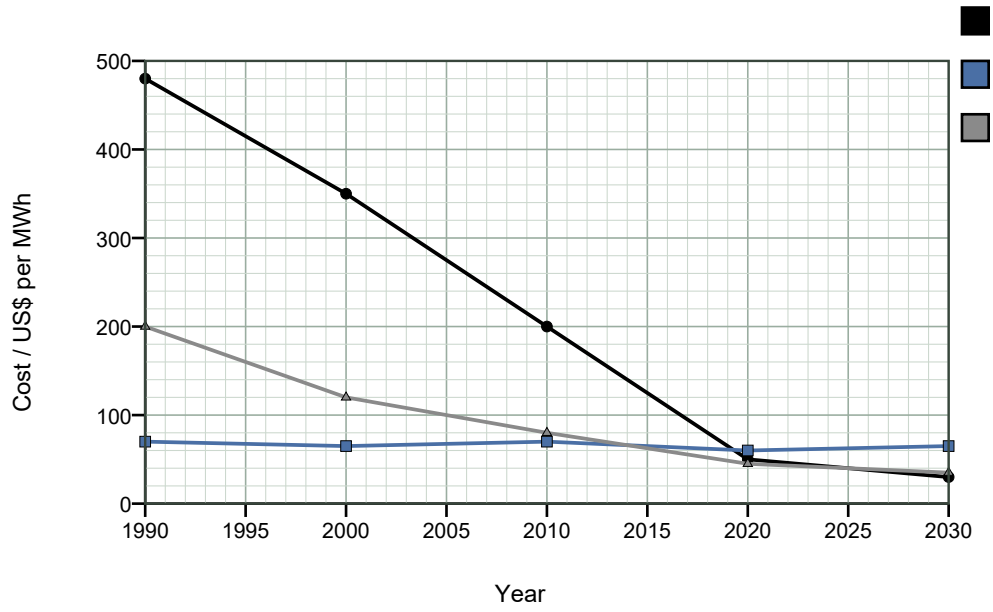
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0 2 Figure 1 shows the estimated cost of generating electricity from three sources between 1990 and 2020, and projections to 2030.

Figure 1



0 2 . 1 Describe the trends shown in Figure 1.

[3 marks]

0 2 . 2 Explain why new energy technologies are expensive when first introduced and why their costs then fall.

[3 marks]

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0 2 . 3 Explain why a renewable energy project may still be at a financial disadvantage compared with a gas-fired power station of the same total cost.

[2 marks]

0 2 . 4 Explain why the price paid for fossil fuel energy does not reflect its full cost.

[2 marks]

10

0 3 . 1 Explain how each of the following human activities alters processes in the hydrological cycle.

[6 marks]

Deforestation

Urban development

Agriculture

0 3 . 2 Describe **two** ways in which global climate change alters the hydrological cycle.

[2 marks]

1 .....

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2 .....

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0 3 . 3 Explain why the impact of industrial and domestic water use on the hydrological cycle is usually local.

[2 marks]

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0 4 . 1 Explain the roles of **three** named groups of soil organisms in maintaining soil fertility.

[6 marks]

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0 4 . 2 Explain why a soil pH between 5.5 and 7.0 is best for fertility.

[2 marks]

0 4 . 3 Explain how soil compaction by heavy machinery reduces fertility.

[2 marks]

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0 5 Table 1 lists some techniques used in ecological monitoring.

0 5 . 1 Complete Table 1 by giving one application and one limitation of each technique.

[6 marks]

Table 1

Technique	Application	Limitation	
Image databases of unique markings			
eDNA sampling			
Geolocator tags			

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0 5 . 2 Explain how satellite radar (microwave) sensors are useful for monitoring oil pollution and sea conditions.

[2 marks]

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0 5 . 3 Give **two** types of indirect evidence of an animal's presence and the information each can give.

[2 marks]

1 .....

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2 .....

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0 6 . 1 Which of the following is the main advantage of composting organic waste rather than landfilling it? Tick **one** box.

[1 mark]

A Aerobic decomposition produces little methane and returns nutrients to the soil ☒

B It produces more electricity ☐

C It removes heavy metals ☐

D It destroys asbestos fibres ☐

0 6 . 2 Explain why the ash from waste incinerators must be disposed of carefully.

[2 marks]

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0 6 . 3 Give **two** features of a well-managed landfill site that reduce pollution.

[2 marks]

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5

0 7 Table 1 compares the energy used to produce 1 tonne of aluminium from bauxite ore with the energy used to produce 1 tonne from recycled aluminium cans.

Table 1

Source	**Energy used / GJ per tonne**	**CO_2 emitted / tonnes per tonne of aluminium**
Bauxite ore (extraction, refining, smelting)	210	12
Recycled cans (collection, sorting, remelting)	12	0.7

0 7 . 1 Calculate the percentage energy saving from recycling aluminium rather than extracting it from ore.

[2 marks]

Answer ..... %

0 7 . 2 Give **three** advantages of recycling metals other than energy saving.

[3 marks]

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3 .....

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0 7 . 3 Explain **four** difficulties that reduce the effectiveness of recycling schemes.

[4 marks]

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0 7 . 4 Explain what is meant by **Cradle to Cradle** design.

[1 mark]

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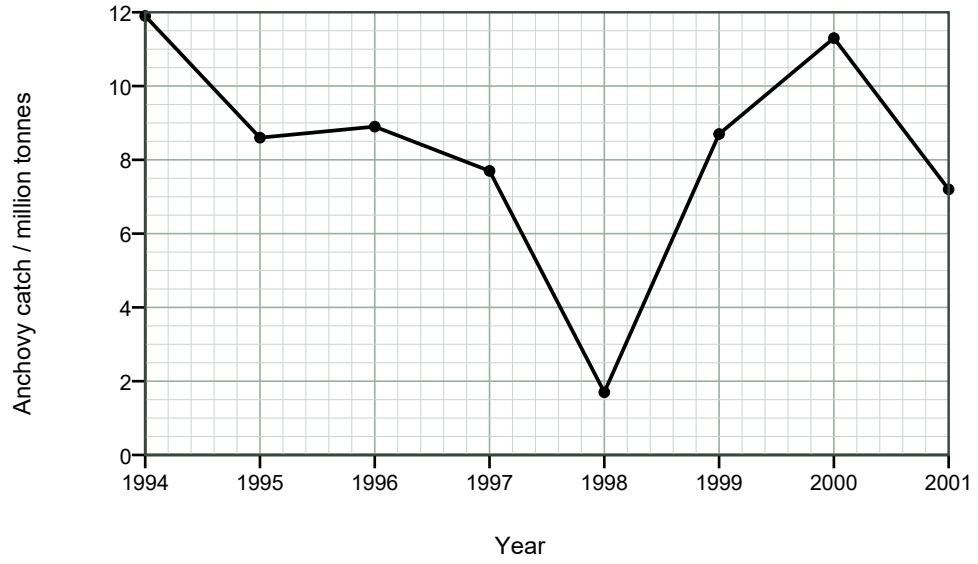
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0 8 Figure 1 shows the sea surface temperature anomaly in the eastern Pacific off Peru and the annual Peruvian anchovy catch for a series of years including a strong El Niño event in 1997-98.

Figure 1



0 8 . 1 Describe the normal (non-El Niño) pattern of winds and ocean currents in the tropical Pacific and explain why the Peruvian coast supports a rich fishery.

[4 marks]

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0 8 . 2 Explain how an El Niño event caused the collapse of the anchovy catch in 1998 shown in Figure 1.

[3 marks]

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0 8 . 3 Describe **two** other effects of El Niño events.

[2 marks]

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2 .....

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0 8 . 4 Suggest why some scientists link global climate change to changes in the frequency of El Niño events, and why this link is uncertain.

[1 mark]

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0 9 . 1 Explain the difference between a **mutagen**, a **carcinogen** and a **teratogen**.

[3 marks]

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0 9 . 2 Explain why a birth abnormality caused by a teratogen cannot be inherited by future generations, whereas one caused by a mutagen may be.

[2 marks]

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0 9 . 3 Distinguish between **somatic** and **gonadic** effects of a mutagen.

[2 marks]

0 9 . 4 Explain what is meant by **synergism** and give an example.

[3 marks]

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1 0 . 1 Explain how the properties of smoke and of the gases involved in acid rain lead to their environmental effects, and evaluate the effectiveness of the methods used to control them.

[9 marks]

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Extra space

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1 0 . 2 State **one** effect of tropospheric ozone on plants.

[1 mark]

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Write an essay on one of the following topics.

11 . 1 Discuss the difficulties in monitoring and predicting global climate change and evaluate the methods used to overcome them.

[25 marks]

OR

11 . 2 Discuss the extent to which energy conservation technologies could reduce the need for new energy supplies in a developed country.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS